



CAT 2026

PRE-FOUNDATION BATCH

Lecture-02

Algebraic Identities



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TOPICS

TO BE COVERED



- Concept & Numerical based on 'Basic Algebraic Identities'
- Concept & Numerical based on 'Special Case'

Basic Algebraic Identities :



① $(a+b)^2 = a^2 + 2ab + b^2$

$(a+b)^2 - 2ab = a^2 + b^2$

② $(a-b)^2 = a^2 - 2ab + b^2$

$(a-b)^2 + 2ab = a^2 + b^2$

③

$$a^2 - b^2 = (a+b)(a-b)$$

Basic Algebraic Identities :



$$(4) \quad (a+b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$$

$$(a+b)^3 = a^3 + b^3 + 3ab[a+b]$$

$$(5) \quad (a-b)^3 = a^3 - 3a^2b + 3ab^2 - b^3$$

$$(a-b)^3 = a^3 - b^3 - 3ab[a-b]$$

$$(6) \quad a^3 + b^3 = (a+b)(a^2 - ab + b^2)$$

$$(7) \quad a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$



Basic Algebraic Identities :

- $(a + b)^2 = a^2 + 2ab + b^2$
- $(a - b)^2 = a^2 - 2ab + b^2$
- $a^2 - b^2 = (a + b)(a - b)$
- $a^2 + b^2 = (a + b)^2 - 2ab$ or $a^2 + b^2 = (a - b)^2 + 2ab$
- $(a + b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$ or $(a + b)^3 = a^3 + b^3 + 3ab(a + b)$
- $(a - b)^3 = a^3 - 3a^2b + 3ab^2 - b^3$ or $(a - b)^3 = a^3 - b^3 - 3ab(a - b)$
- $a^3 + b^3 = (a + b)(a^2 - ab + b^2)$
- $a^3 - b^3 = (a - b)(a^2 + ab + b^2)$



Question No. 1



#Q. If $x^2 - y^2 = 80$ and $x - y = 8$, then the average of x and y is

$$(x + y)(\underline{x - y}) = 80$$

$$(x + y) 8 = 80$$

$$x + y = 10$$

Avg of x & y

$$= \frac{(x + y)}{2}$$

$$= \frac{10}{2} = 5$$

A 1

B 2

C 5

D 4



Question No. 2

$$(x y)^m = x^m y^m$$



#Q. If $a = 23$ and $b = -29$, then the value of $(\underline{25a^2} + \underline{40ab} + \underline{16b^2})$ is

A

1 ✓

B

-1

C

0

D

2

$$\begin{aligned} (x+y)^2 &= x^2 + 2xy + y^2 \\ &= (5a)^2 + 2(5a)(4b) + (4b)^2 \\ &= [5a + 4b]^2 \\ &= [5(23) + 4(-29)]^2 \\ &= [115 - 116]^2 \\ &= (-1)^2 = 1 \end{aligned}$$



Question No. 3



#Q. If $p + q = 10$ and $pq = 5$ then the numerical value of $\frac{p}{q} + \frac{q}{p}$ will be

$$p^2 + q^2 = (p + q)^2 - 2pq$$

~~$$p^2 - q^2 = (p - q)^2 + 2pq$$~~

A 16

B 18

C 20

D 22

$$\begin{aligned} & \frac{p}{q} + \frac{q}{p} \\ &= \frac{p^2 + q^2}{pq} \\ &= \frac{(p + q)^2 - 2pq}{pq} \\ &= \frac{(10)^2 - 2(5)}{5} = \frac{90}{5} = 18 \end{aligned}$$



Question No. 4



#Q. If $P = 101$, then the value of $\sqrt[3]{P(p^2 - 3p + 3) - 1}$ is :

$$\sqrt[3]{p^3 - 3p^2 + 3p - 1}$$

$$\sqrt[3]{p^3 - 3p^2(1) + 3p(1)^2 - 1^3}$$
$$\boxed{a^3 - 3a^2b + 3ab^2 - b^3} \rightarrow (a-b)^3$$

$$\sqrt[3]{(p-1)^3} = p-1 = 101-1 = 100$$

A

100

B

101

C

102

D

1000



Question No. 5

$$(a-b)^3 = a^3 - 3a^2b + 3ab^2 - b^3$$



#Q. If $x = \sqrt[3]{5} + 2$, then the value of $x^3 - 6x^2 + 12x - 13$ is :

A

-1

B

0 ✓

C

2

D

1

$$x = \sqrt[3]{5} + 2$$

$$(x-2) = \sqrt[3]{5}$$

$$(x-2)^3 = (\sqrt[3]{5})^3$$

$$x^3 - 3x^2(2) + 3x(2)^2 - 2^3 = 5$$

$$x^3 - 6x^2 + 12x - 8 - 5 = 0$$

$$x^3 - 6x^2 + 12x - 13 = 0$$



Question No. 6



#Q. $\frac{2125 \times 2125 \times 2125 + 875 \times 875 \times 875}{2125 \times 2125 - 2125 \times 875 + 875 \times 875} = ?$

$2125 \rightarrow a$
 $875 \rightarrow b$

$$\frac{a^3 + b^3}{a^2 - ab + b^2}$$
$$\frac{(a+b) \cancel{(a^2 - ab + b^2)}}{\cancel{(a^2 - ab + b^2)}}$$

$$a + b = 2125 + 875 = \underline{\underline{3000}}$$

A 3000

B 2000

C 4000

D 1000



Question No. 7

$$(a b)^m = a^m b^m$$
$$\left(\frac{a}{b}\right)^m = \frac{a^m}{b^m}$$



#Q. If $xy(x + y) = 1$, then the value of $\frac{1}{x^3 y^3} - x^3 - y^3$ is :

A -1

B 3 ✓

C -3

D 1

$$xy(x + y) = 1$$

$$x^2 y + xy^2 = \textcircled{1}$$

$$xy.(x + y) = 1$$

$$x + y = \frac{1}{xy}$$

$$(x + y)^3 = \left(\frac{1}{xy}\right)^3$$

$$x^3 + \boxed{3x^2 y + 3xy^2} + y^3 = \frac{1}{x^3 y^3}$$

$$3x^2 y + 3xy^2 = \frac{1}{x^3 y^3} - x^3 - y^3$$

$$3(\underline{x^2 y + xy^2}) = \frac{1}{x^3 y^3} - x^3 - y^3$$

$$\boxed{3 = \frac{1}{x^3 y^3} - x^3 - y^3}$$



Question No. 8

$$(a+b)^2 = a^2 + 2ab + b^2$$



#Q. If $x^2 - \sqrt{7}x + 1 = 0$, then what is the value of $x^2 + \frac{1}{x^2}$

A

7

B

5

C

9

D

11

$$x^2 + 1 = \sqrt{7}x$$

$$\frac{x^2}{x} + \frac{1}{x} = \frac{\sqrt{7}x}{x}$$

$$\left(x + \frac{1}{x}\right)^2 = (\sqrt{7})^2$$

$$x^2 + 2\left(\frac{1}{x}\right)x + \left(\frac{1}{x}\right)^2 = 7$$

$$x^2 + 2 + \frac{1}{x^2} = 7$$

$$x^2 + \frac{1}{x^2} = 5$$



| Special Case :

$$0^2 + 0^2 + 0^2 = 0$$

If $a^2 + b^2 + c^2 = 0$
Then $a = b = c = 0$

$$a^4 + b^4 + c^4 = 0$$
$$a = b = c = 0$$

$$a^2 + b^2 = 0$$
$$a = b = 0$$

$$a^2 + b^2 + c^2 + d^2 = 0$$
$$a = b = c = d = 0$$

$$a^2 + b^2 + c^2 = 0$$
$$a = b = c = 0$$



Question No. 9



#Q. If $(a - 3)^2 + (b - 4)^2 + (c - 9)^2 = 0$, then the value of $\sqrt{a + b + c}$ is:

A

5

B

3

C

4

D

6

$$\begin{array}{c|c|c} a-3=0 & b-4=0 & c-9=0 \\ \hline \boxed{a=3} & \boxed{b=4} & \boxed{c=9} \end{array}$$

$$\sqrt{a+b+c} = \sqrt{3+4+9} = \sqrt{16} = 4$$



Question No. 10



#Q. If $(x + y - z - 1)^2 + (z + x - y - 2)^2 + (z + y - x - 4)^2 = 0$,
then find $x + y + z$?

A

5

B

3

C

4

D

7

$$\begin{array}{l|l|l} x+y-z-1=0 & z+x-y-2=0 & z+y-x-4=0 \\ \hline x+y-z=1 & z+x-y=2 & z+y-x=4 \end{array}$$
$$\cancel{x+y-z} + \cancel{z+x-y} + \cancel{z+y-x} = 1 + 2 + 4$$
$$\boxed{x+y+z=7}$$



Question No. 11



#Q. If $\underline{x}^2 + \underline{y}^2 - \underline{4x} - \underline{4y} + \underline{8} = 0$, then the value of $(x - y)$ is:

$$a^2 - 2ab + b^2 = (a - b)^2$$

$$(x^2 - 4x + 4) + (y^2 - 4y + 4) = 0$$

$$x^2 - 2(x)(2) + 2^2 + y^2 - 2(y)(2) + 2^2 = 0$$

$$(\underline{x-2})^2 + (\underline{y-2})^2 = 0$$

$$\therefore \begin{array}{l|l} x-2=0 & y-2=0 \\ x=2 & y=2 \end{array}$$

A

5

B

8

C

4

D

0 ✓

2 Mins Summary



- Concept & Numerical based on 'Basic Algebraic Identities'
- Concept & Numerical based on 'Special Case'



Thank
You